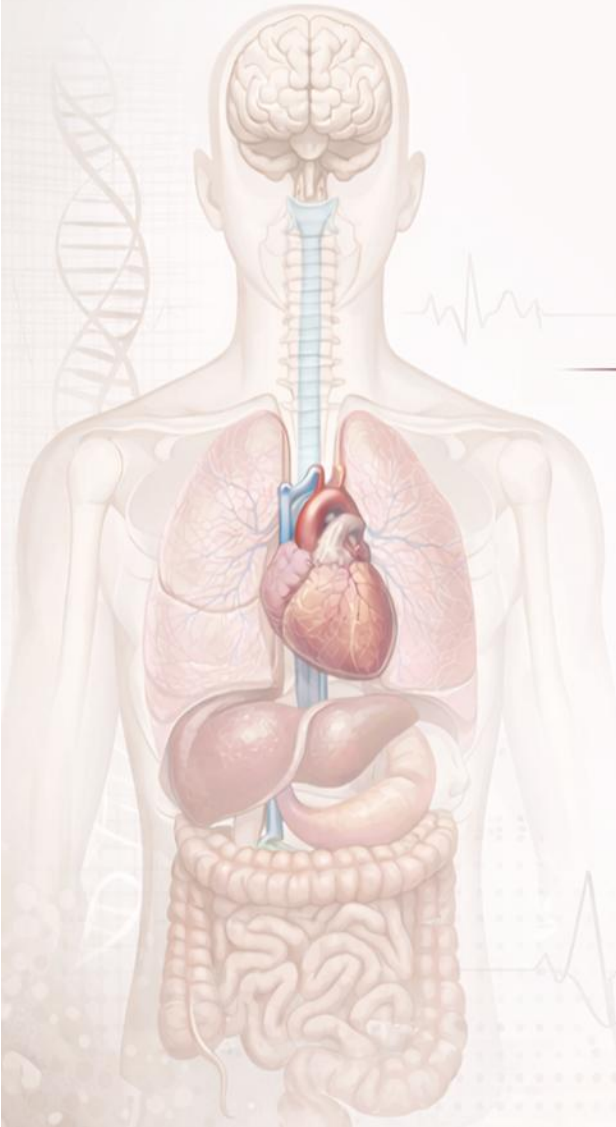
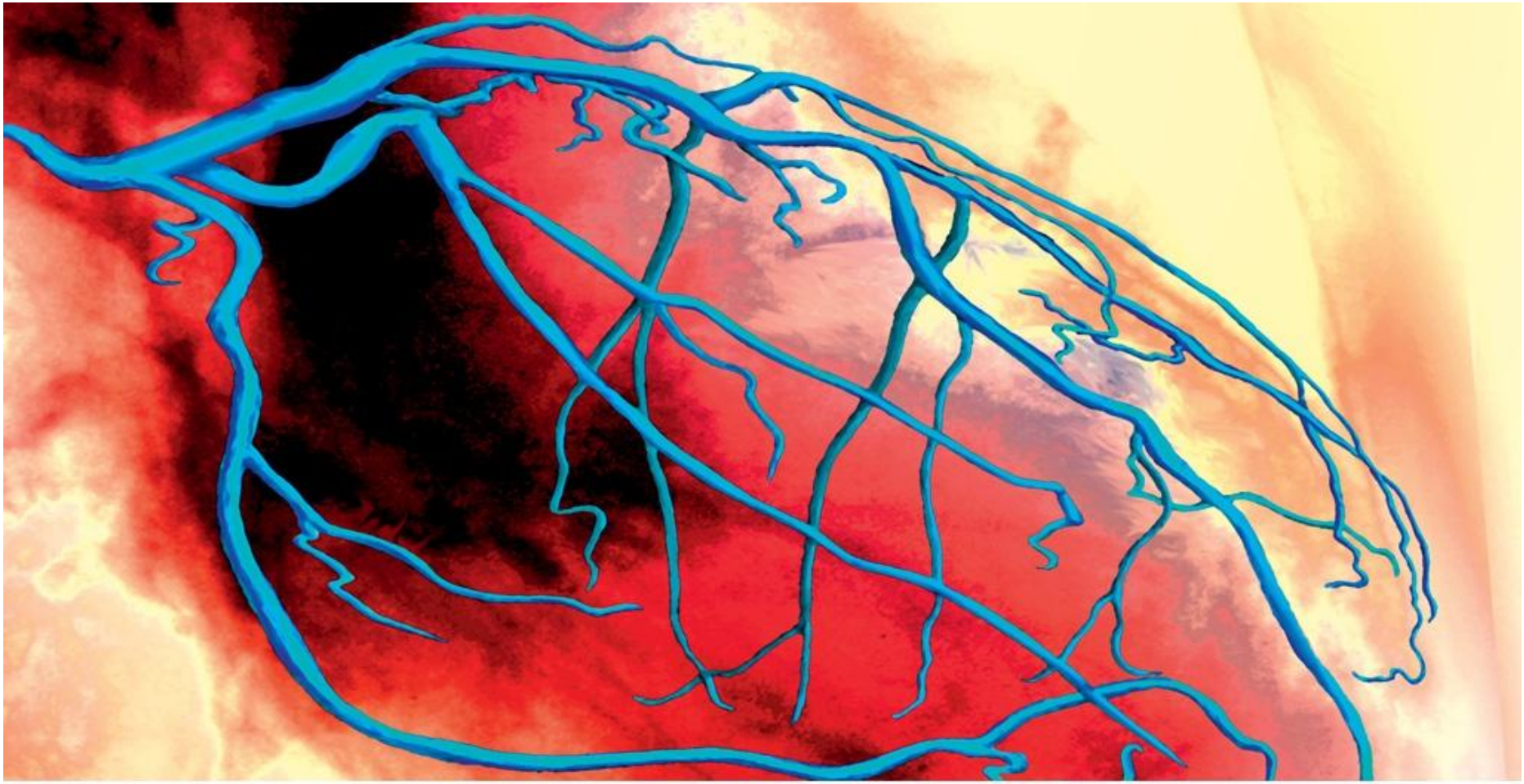


Chapter 9.
Cardiac Physiology

KOREA UNIVERSITY
1905

A colored X-ray angiogram of healthy heart



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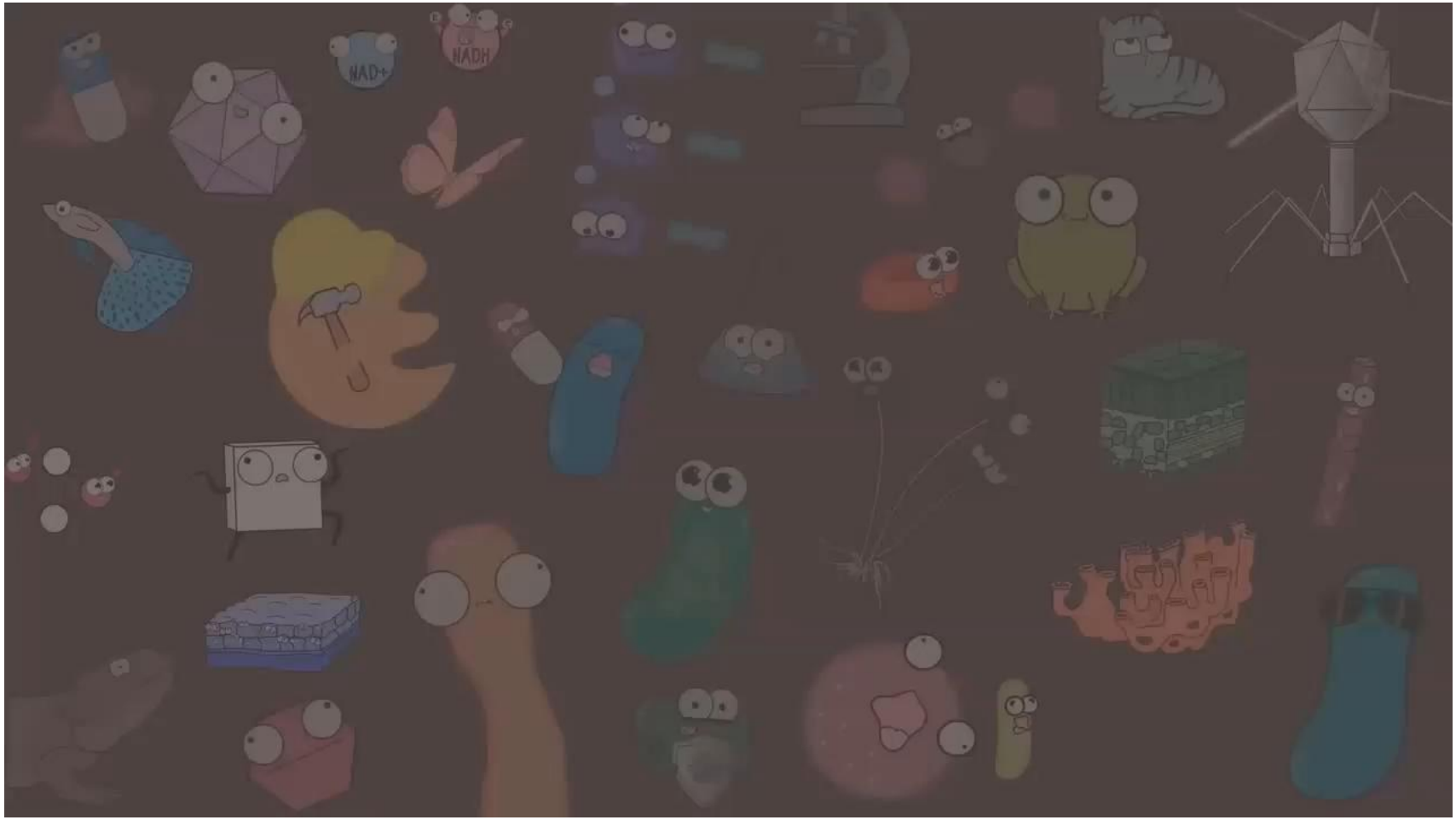
Circulatory system in terms of homeostasis

- To maintain homeostasis, essential materials like O_2 and nutrients must be obtained from the exterior and delivered to the cells
- Waste products must be continuously removed, too
- Heat generated by muscles must be transported to skin to maintain body temperature
- Transfer of hormones, secreted by endocrine system to deliver regulatory chemical messengers to their site of action

- Body's transport system: heart, blood vessels and blood

9.1 Anatomy of the Heart

Circulatory system and pathway of blood through heart



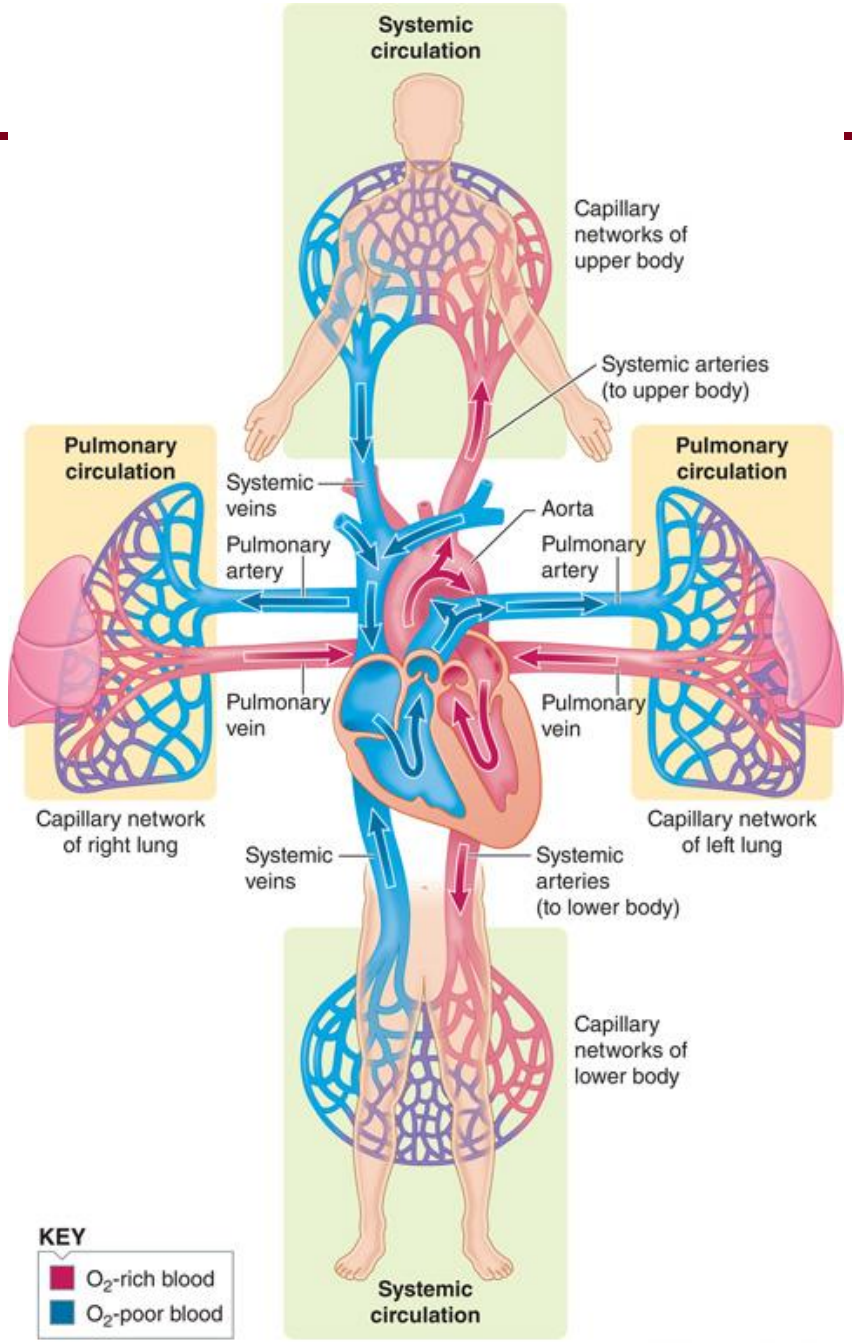
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Circulatory system

- Three basic components
- Heart: **pump** that transmits pressure to the blood to establish the pressure gradient need for their flow into tissues
  - Blood flows down a pressure gradient, from high to low
- Blood vessels: **passageways** where blood is directed and distributed from the heart to all body parts and finally returned to the heart
  - Smallest blood vessels (capillaries) are designed for rapid material exchange with the surrounding tissue
- Blood: transport **medium** within which materials being transported long distances, such as O_2, CO_2, nutrients, wastes, electrolytes, and hormones are dissolved or suspended

Circulatory system

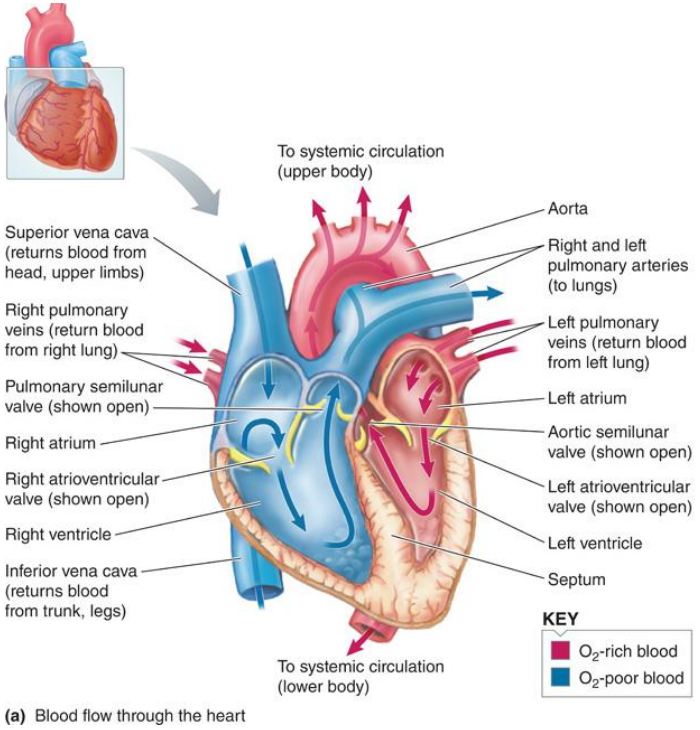
- Blood travel
- Pulmonary circulation
  - Closed loop of vessels carrying blood between heart and lungs
  - Two directions: looping through the right and left lung
- Systemic circulation
  - Circuit of vessels carrying blood between heart and all body system, except for lungs
  - Two directions: looping through the upper and lower halves
- The heart is positioned in the middle of thoracic (chest) cavity
  - Between sternum (흉골) in the front and vertebrae (척추) in the back



The heart

- Base (심장바닥) and apex (심첨)

- The heart is a dual pump
- Anatomically divided into the right and left halves as two separate pumps
- Two chambers (upper and lower) for each half
  - Upper chamber: **atrium** (심방) receive blood returning to heart via **veins** and transfer the blood to ventricle
  - Lower chamber: **ventricle** (심실) pump blood from the heart through **arteries**
  - Two chambers are separated by **septum** (사이막, 중격), muscular partition preventing blood mixing
  - Right side receives and pumps **O_2-poor blood** and left side receives and pumps **O_2-rich blood**

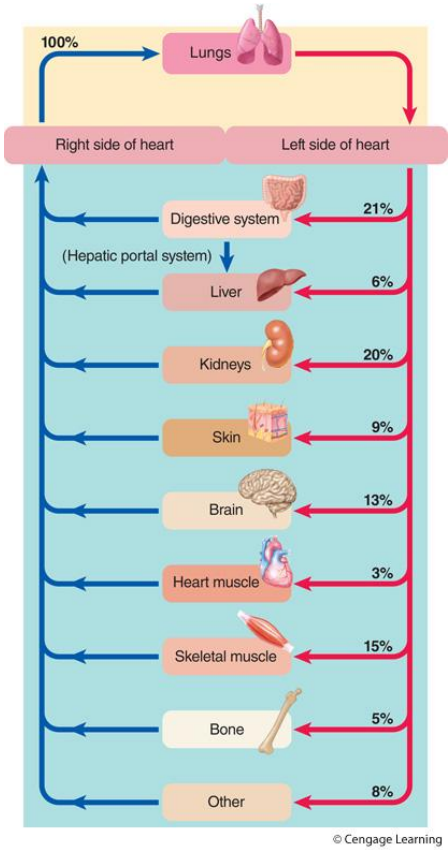


(a) Blood flow through the heart

- CPR (심폐소생술) rhythmically compresses the sternum so that blood is squeezed out into the blood vessels and tissues

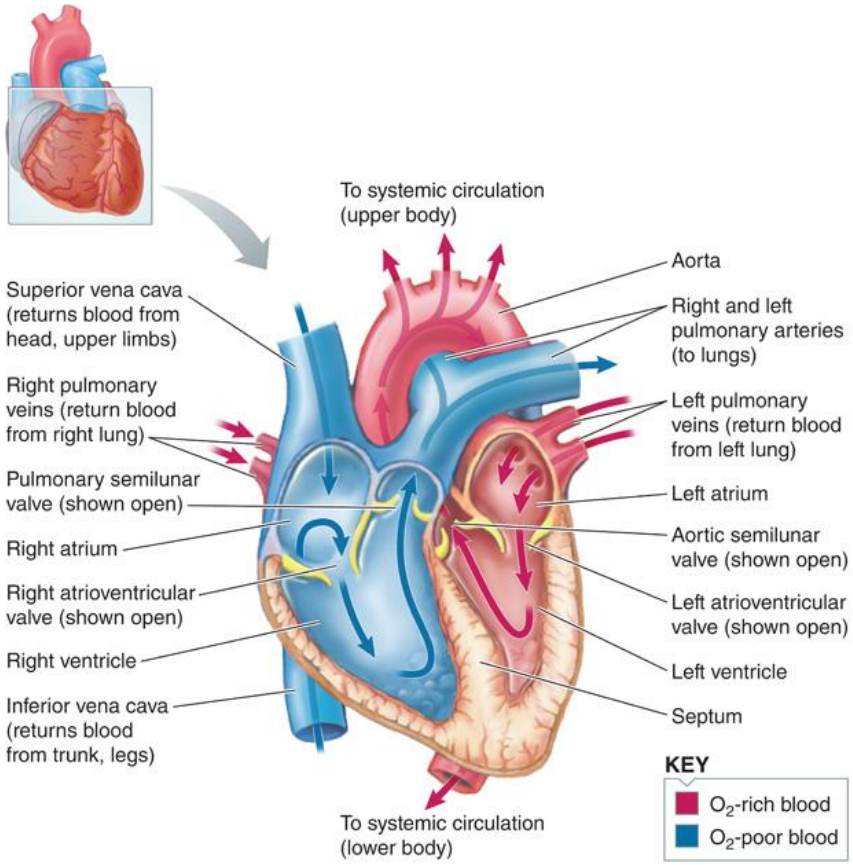
The complete circuit of blood flow

- **Vena cava**: two large veins returning to right atrium
  - Superior vena cava (상대정맥): returns blood from upper level
  - Inferior vena cava (하대정맥): returns blood from lower level
  - Contains O_2-poor and CO_2-rich blood
- **Pulmonary artery** (폐동맥)
  - Partially deoxygenated blood is pumped out
  - The artery forms two branches, to each of two lungs
- Lungs
  - The blood loses extra CO_2 and picks up fresh O_2 via gas exchange with the air sacs of lungs
  - The fresh blood returns to left atrium via pulmonary veins
- **Aorta** (대동맥)
  - Single artery carrying blood away from the left ventricle
  - Major arteries branch from aorta to bring blood to various organs
- Cardiac output
  - Part of blood is delivered in parallel to many body parts including cardiac muscle and lung (except for air sacs)
  - Fresh blood supply to each body part
  - Tissue cells take O_2 use it to oxidize nutrients for energy production and form CO_2 as a waste product

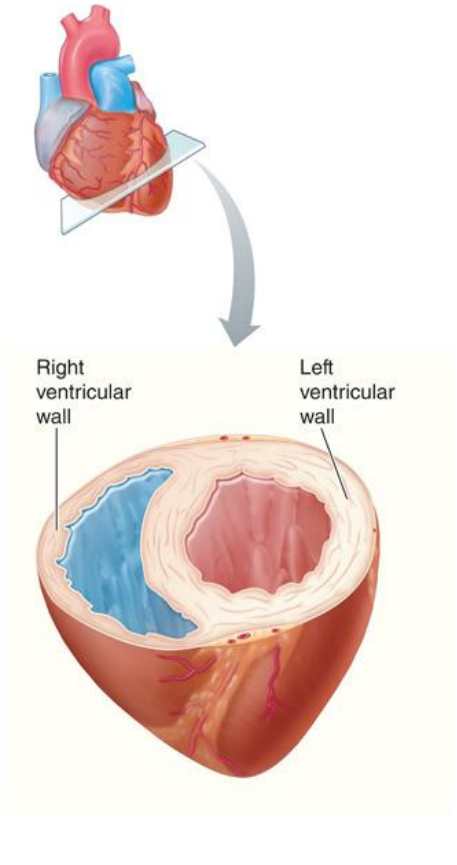


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Comparison of the right and left pumps



(a) Blood flow through the heart

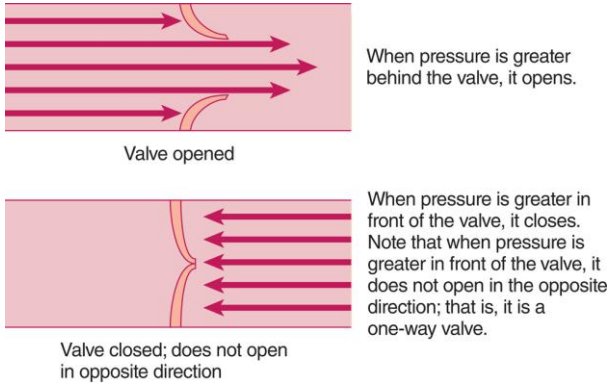


(b) Thickness of right and left ventricles

Both sides of the heart simultaneously pump equal amounts of blood
Pulmonary system (low-pressure, low-resistance) vs systemic system (high-pressure, high-resistance)
Left ventricle muscle is thicker than that on the right side, making left side a stronger pump

Heart valves

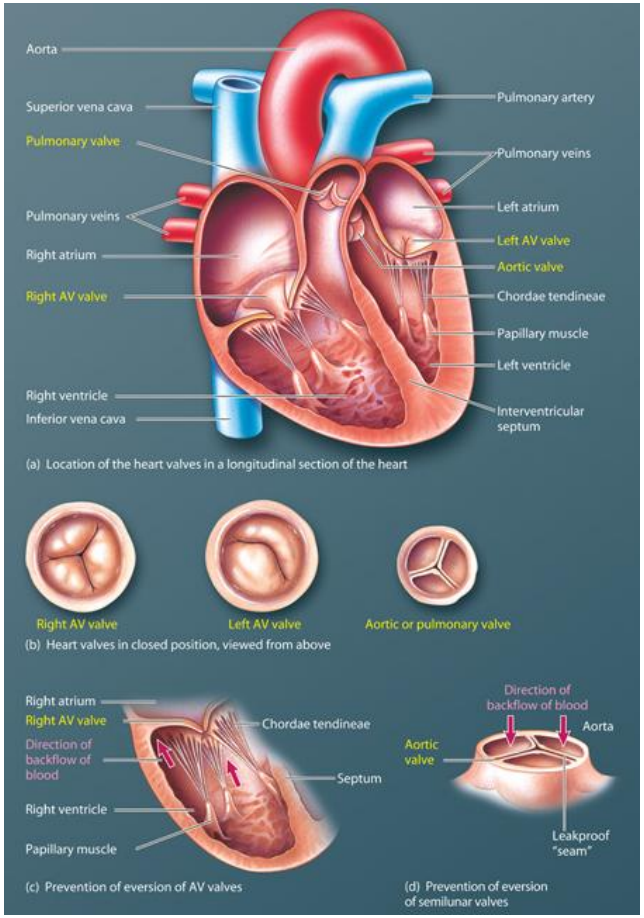
- Blood flows through the heart in one fixed direction
- **Four** one-way **heart valves** ensure unidirectional blood flow
- Valves open and close passively because of pressure difference



- Two atrioventricular (**AV valves**) between atrium and ventricle
  - Let blood flow from the atria during ventricular filling, when atrial pressure exceeds ventricular pressure
  - Prevent backflow of blood during ventricular emptying, when atrial pressure is greatly smaller than ventricular pressure
  - Enables efficient pumping into the arteries
  - Right AV valve = tricuspid valve (three leaflets)
  - Left AV valve = bicuspid valve (two leaflets) or mitral valve

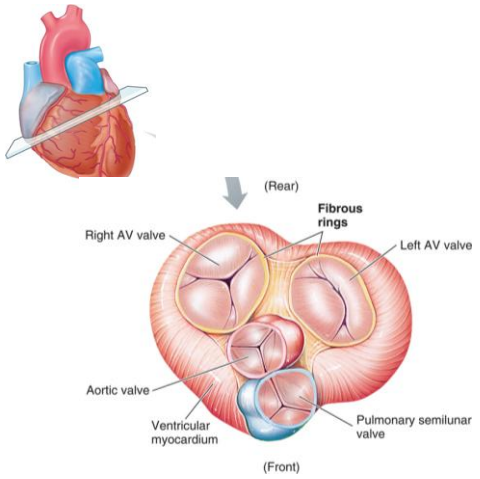
Heart valves

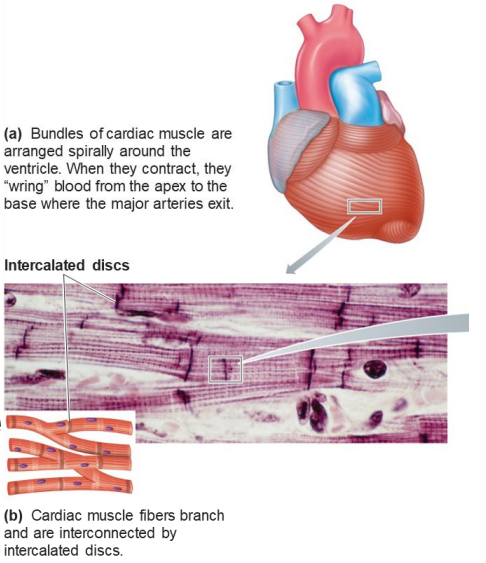
- Chordae tendineae
  - Tough, thin cords of tendinous-type tissue
  - Prevent the AV valve from everting
  - Extend from each cusp and attach to papillary muscles protruding from ventricular wall's surface
- Two **semilunar valves** between ventricle and arteries
  - Half-moon-shaped pocket with three cusps
  - Aortic valve (left)
  - Pulmonary valve (right)
  - These valves are forced open when the left and *right* ventricular pressures exceed the pressure in aorta or *pulmonary* artery
  - Their closure results when the ventricles relax and ventricular pressure falls below the arterial pressures, preventing artery-to-ventricle blood flow
- No valves between veins and atria



Heart valves and heart walls

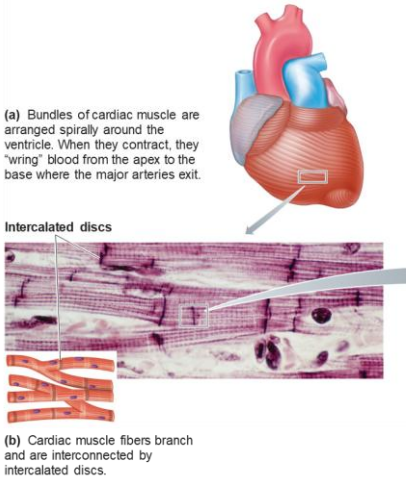
- **Fibrous skeleton** (섬유골격) of the valves
  - Interconnected four fibrous rings surrounding and supporting four heart valves
  - Separates the atria from ventricle
  - Provide a fairly rigid structure for attachment of cardiac muscle
- The heart walls are composed of spirally arranged cardiac muscle fibers
  - The spiral arrangement enables a twisting motion of the heart, which enhances the efficiency of blood ejection during contraction
  - This structure allows heart to generate greater force with less energy and facilitates efficient filling during relaxation by promoting rapid untwisting and suction
- Three layers of heart wall
  - Endothelium: thin, inner, epithelial tissue layer lining the entire circulatory system
  - **Myocardium**: thick, middle layer composed of **cardiac muscle**
  - Epicardium: thin, outer layer covering the heart

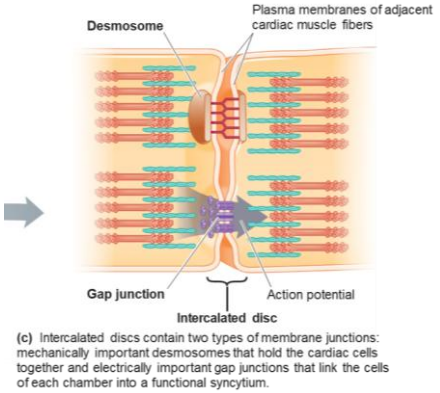




Cardiac muscle fibers

- Myocardium consists of bundles of cardiac muscle fibers arranged spirally around the heart perimeter
- Wringing effect: ventricular muscle contraction moves the apex upward and exerts pressure on the blood
  - This rhythmic contractile activity of cardiac muscle is supported by energy-generating mitochondria, and each fiber receives rich blood supply from one capillary
- Cardiac muscle fibers are interconnected by **intercalated discs** and form functional syncytium
  - Intercalated disc: adjacent cells are joined end-to-end at here
  - **Desmosomes**: mechanically holds cells together, abundant in tissues subject to huge mechanical stress
  - **Gap junctions**: two membranes adhere to form gap junction, areas of low electrical resistance allowing AP spread one-to-one
  - Some specialized cardiac muscle cells can generate AP -> excited and contract as a single functional syncytium
  - Two atria and two ventricles each form a functional syncytium
  - No gap junctions joint the atrial and ventricular muscle cells, they are electrically separated by **nonconductive** fibrous skeleton
  - However, there is a conducting system relaying electrical excitation from A to V, ensuring synchronization





9.2 Electrical Activity of the Heart

Electrical activity of the heart

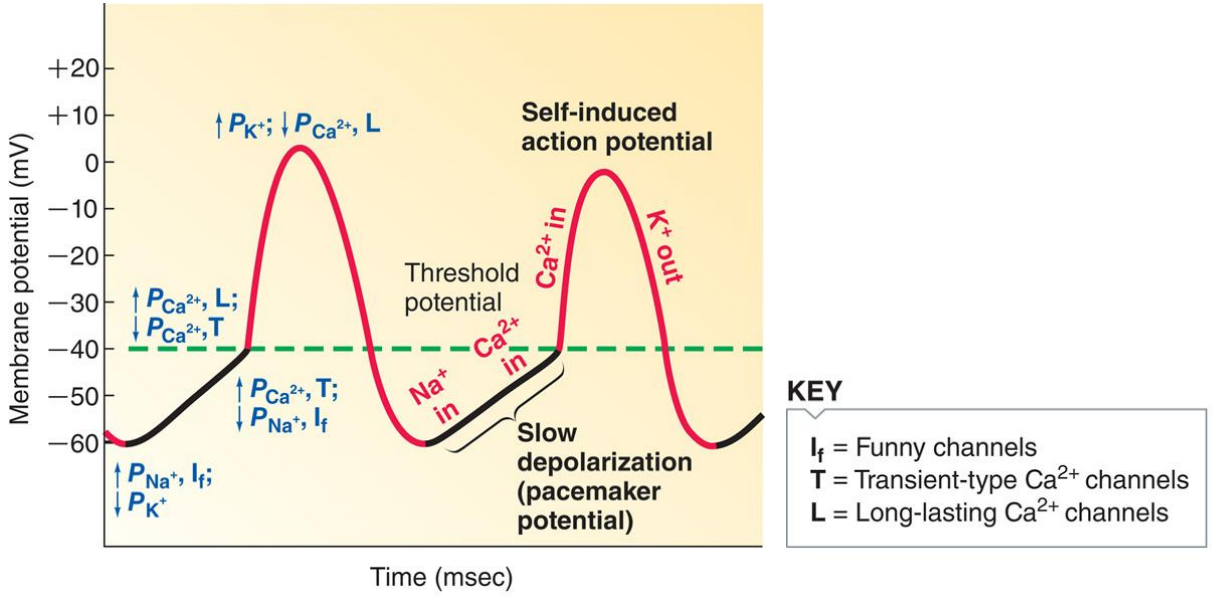
- Cardiac muscle cell contraction is triggered by APs sweeping across the muscle cell membranes
- Autorhythmicity or automaticity: rhythmic contraction of the heart as a result of **self-generated action potentials**

- Two specialized cardiac muscle cell types
- **1) Contractile cells**
- 99% of the cardiac muscle cells doing the mechanical work of pumping
- Do not initiate, but just relay APs

- **2) Autorhythmic cells**
- Initiate and conduct the APs making contraction of contractile cells
- Do not contract

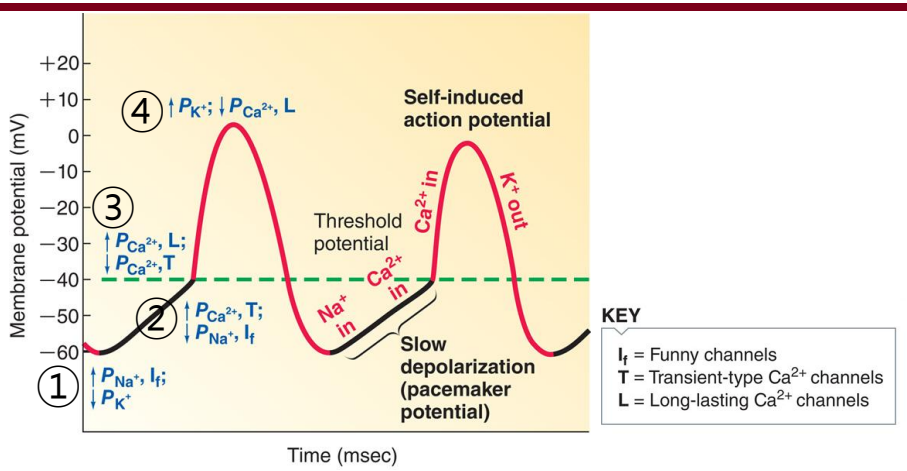
Cardiac autorhythmic cells

- Cardiac autorhythmic cells display pacemaker activity
  - Unlike nerve and skeletal muscle cells at rest, cardiac autorhythmic cells do not have a resting potential
  - Their membrane potential slowly depolarizes between APs until threshold is reached
- **Pacemaker potential**
  - Autorhythmic cell membrane's slow drift to threshold to cyclically initiate APs
  - APs spread and trigger rhythmic beating without nerve innervation



Cardiac autorhythmic cells – action potential

- **Pacemaker potential**
  - Increased inward Na^+ current
  - Decreased outward K^+ current
  - Increased inward Ca^{2+} current



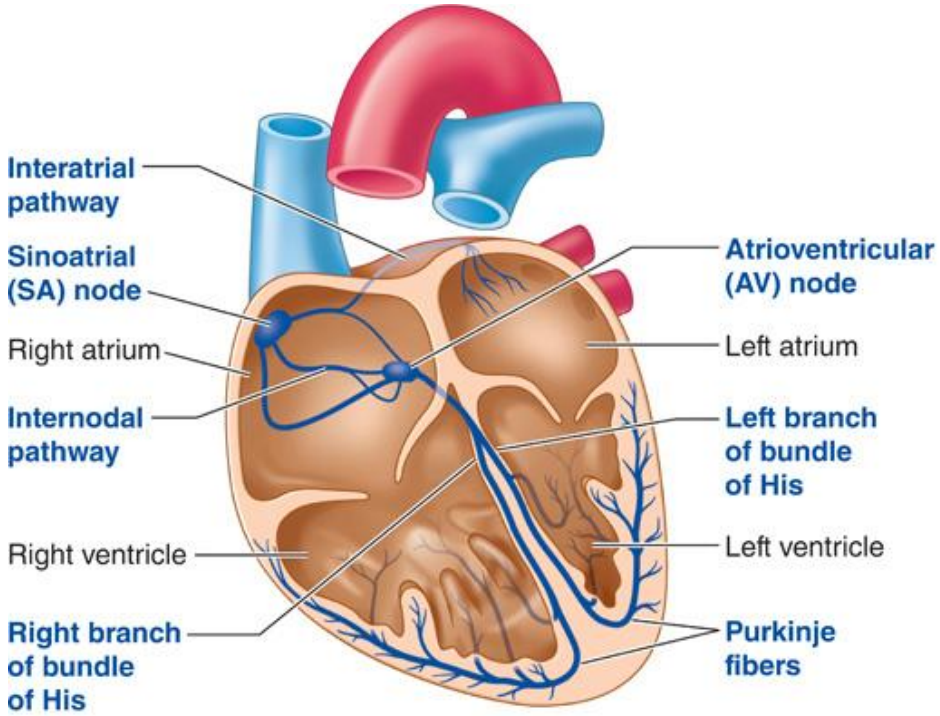
- **Membrane clock mechanism**
- ① Funny channel (I_f channel)
  - Voltage-gated Na^+ channel that opens upon hyperpolarization (more negative)
  - Found only on cardiac pacemaker cells
- ① Delayed K^+ channel closing
  - Voltage-gated K^+ channel opened at depolarized potentials and gradually close during hyperpolarization
  - Decreasing K^+ efflux contributes to slow depolarization and return membrane potential toward threshold
- ② Transient (T-type) Ca^{2+} channel opening
  - Open at the timing of funny channel closing, before reaching threshold
  - Brief influx of Ca^{2+} bring the membrane potential to the threshold
  - Then T-type Ca^{2+} channel closes

Cardiac autorhythmic cells – action potential

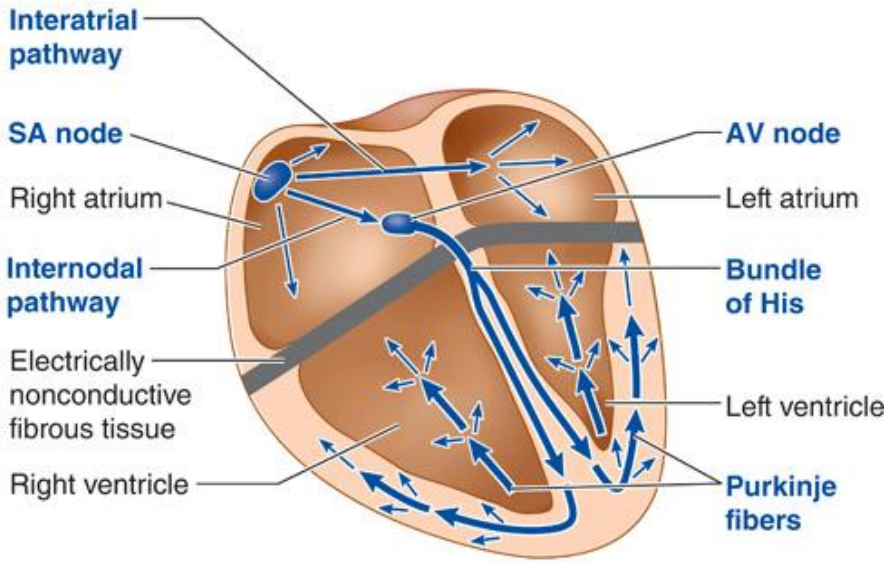
- Beyond the membrane clock mechanism
  - Pacemaker activity is not driven by membrane currents alone
  - Intracellular Ca^{2+} cycling (Ca^{2+} clock mechanism) also contributes
  - Modern view: coupled-clock system integrates both mechanisms

- **Action potential** beyond pacemaker potential in autorhythmic cells
- ③ L-type Ca^{2+} channel
  - Once threshold is reached, AP opens long-lasting voltage-gated Ca^{2+} channel
  - Large influx of Ca^{2+}, rather than Na^{+}
- ④ Voltage-gated K^{+} channel
  - Opening to induce K^{+} efflux, coupled with the closure of L-type Ca^{2+} channel
- Termination and recycling of action potential
  - Ca^{2+} removal mechanisms
    1) SERCA pump (cytosol → SR)
    2) Na^{+}/Ca^{2+} exchanger (NCX) (Ca^{2+} out / Na^{+} in)
  - Na^{+}/K^{+} ATPase restores ionic gradients
  - Hyperpolarization activates If channels → next cycle begins

Pacemaker cells in the heart



(a) Specialized conduction system of the heart



(b) Spread of cardiac excitation

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Pacemaker cells in the heart

- Special sites capable of autorhythmicity
  - SA node, AV node, Bundle of His (AV bundle), Purkinje fibers

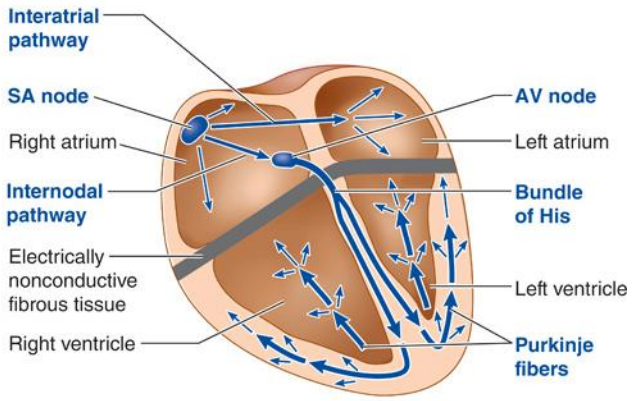
- **SA node** is the *normal* pacemaker of the heart
  - Small region in the right atrial wall near the opening of superior vena cava
- Atrioventricular (**AV node**)
  - Small bundle of cardiac muscle fibers at the base of the right atrium near the septum, just above the junction of atria and ventricle
- **Bundle of His**
  - A tract of cells originating from the AV node and enters the septum
  - Curve around the tip of the ventricular chambers and travel back toward the atria
- **Purkinje fibers**
  - Small terminal fibers extending from the bundle of His and spreading throughout the ventricular myocardium

Pacemaker activity

- Different autorhythmic cells have different rates of slow depolarization to threshold
- Also, the number of APs per minute (rates) each cell type can generate is different
  - SA node: 70-80 APs / min
  - AV node: 40-60 APs / min
  - Bundle of His and Purkinje fibers: 20-40 APs / min
- Once action potential occurs in any cardiac muscle cell, AP is propagated throughout the heart -> **SA node** is the pacemaker of the heart
- If SA node breaks down, AV node takes over pacemaker activity
- Artificial pacemaker when heart rate is abnormally low

Cardiac excitation and muscle contraction

- The spread of cardiac excitation is coordinate
  - Atrial excitation and contraction should be com ventricular contraction
  - Excitation of cardiac muscle fiber should be co chamber contracts as a unit to pump efficiently
  - Two atria and two ventricles are coordinated fo

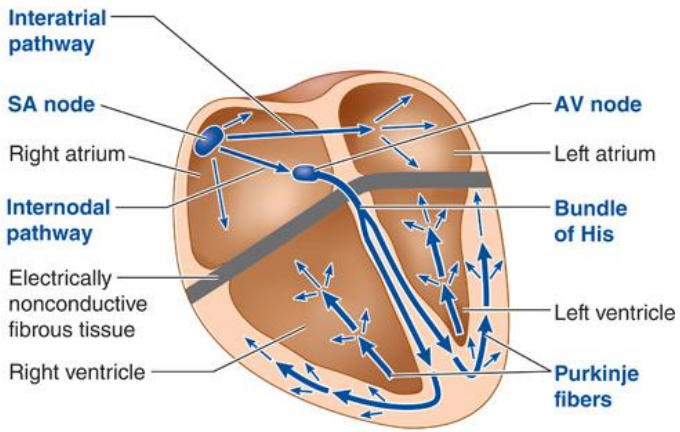


(b) Spread of cardiac excitation

- Once initiated in the SA node, AP spreads throughout the rest of heart
- 1) **Atrial excitation**
  - AP from SA node first spreads throughout both atria, from cell to cell via **gap junctions**
  - Additionally, specialized conduction pathways speed up conduction
  - Interatrial pathway: to left atrium -> both atria contract simultaneously
  - Internodal pathway: to AV node
    - AV node is the only electrical contact point between atria and ventricles
    - Sequential contraction of ventricles after atrial contraction

Cardiac excitation and muscle contraction

- Once initiated in the SA node, AP spreads throughout the rest of heart
- 2) **Conduction between atrium and ventricle**
  - AV nodal delay: AP is conducted relatively slowly through the AV node
  - Slowed conduction between atria and ventricles (~100 ms)
  - Ensures ventricular filling after atrial contraction is completed
- 3) **Ventricular excitation**
  - Bundle of His and Purkinje fibers (ventricular conduction system) travel down the septum and throughout the ventricular myocardium
  - This network of fibers rapidly transmit AP within 30 ms, ensuring that the ventricles contract as a unit

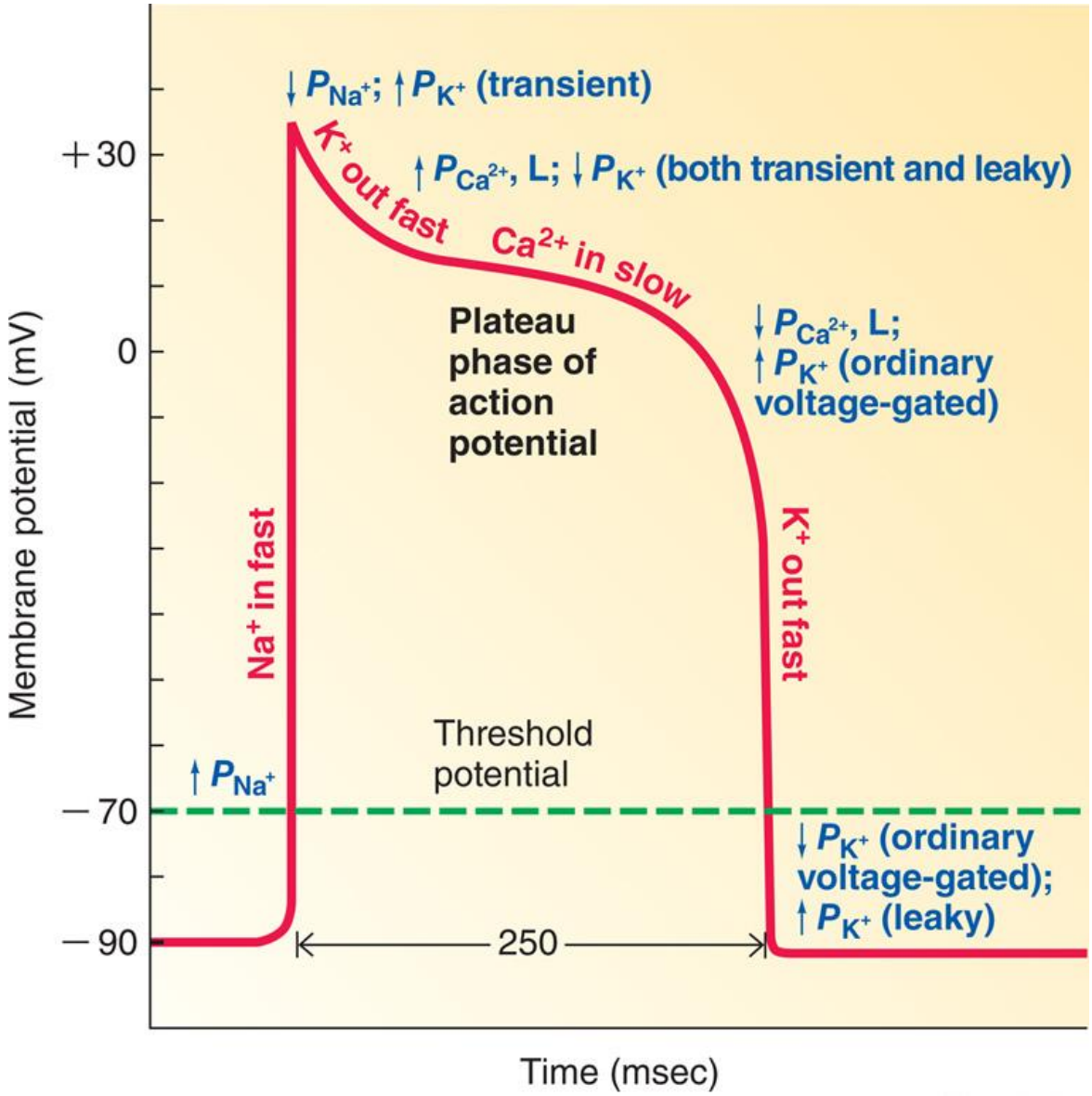


(b) Spread of cardiac excitation

The action potential of cardiac contractile cells

- The APs generated by SA node autorhythmic cells and contractile cells are very different
- Unlike SA node, contractile cells have resting m.p. around -90 mV
  - Very leaky K^+ channel -> K^+ efflux
- Current flow via gap junctions depolarizes the membrane to the threshold, activating AP as follows
  - Activation of voltage-gated Na^+ channel: fast rise in m.p. upto +30 mV
  - Subclass of transient and leaky K^+ channels open: K^+ efflux, brief repolarization (less positive)
  - Uniquely, membrane potential maintains for several hundred ms (**plateau phase** of action potential), maintained by two mechanism
    - Activation of **slow L-type Ca^{2+} channel**: slow and prolonged Ca^{2+} influx
    - Marked decrease in K^+ membrane permeability: reduced K^+ efflux due to channel closure, maintain the depolarization
  - Inactivation (closure) of L-type Ca^{2+} channel and activation of voltage-gated K^+ channel
- Long refractory period prevents tetanus

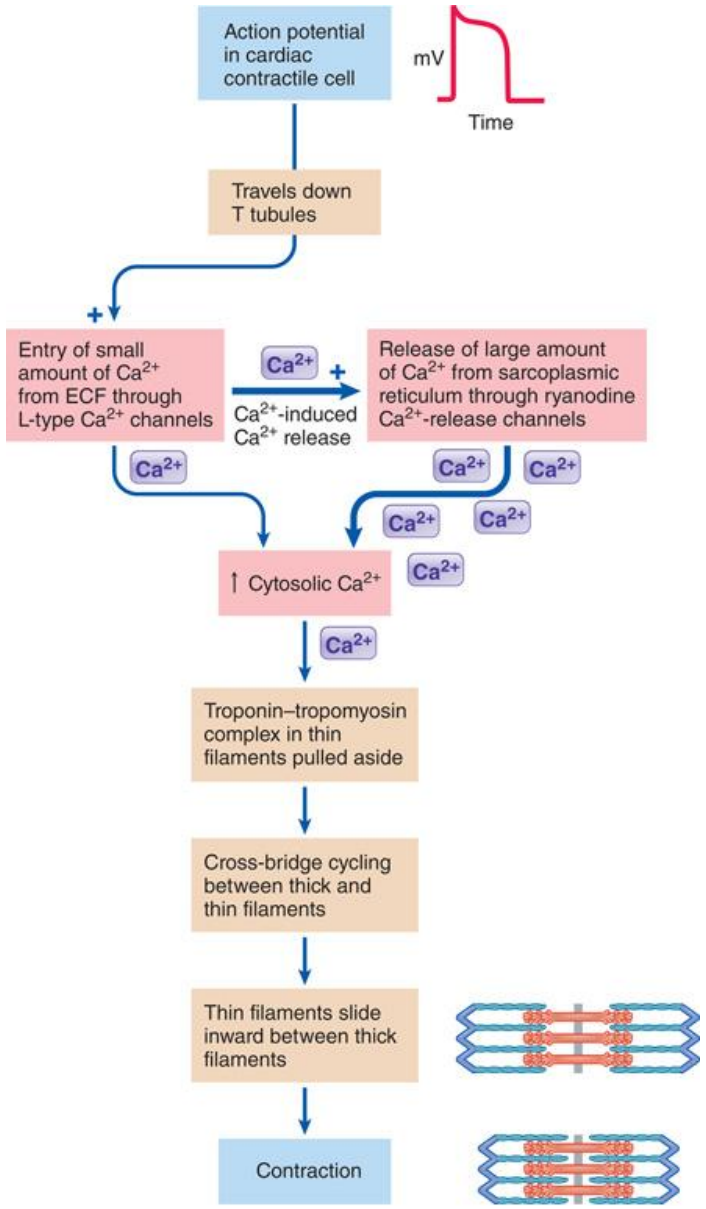
The action potential of cardiac contractile cells



Calcium entry from ECF induces Ca^{2+} release from SR

- L-type Ca^{2+} channels lie mostly in transverse (T) tubules
- Unlike skeletal muscle, Ca^{2+} diffuses into the cytosol from the ECF across the T tubule during AP
- This entering Ca^{2+} triggers the opening of ryanodine Ca^{2+}-release channels in the lateral sac of SR

- **Ca^{2+}-induced Ca^{2+} release**
  - Ca^{2+} entering the cytosol from the ECF induces much larger release of Ca^{2+} into the cytosol from SR (inside the cell)
- Ca^{2+} binds to troponin-tropomyosin complex and pull it aside for cross-bridge
- The amount of Ca^{2+} is more regulated than in skeletal muscle



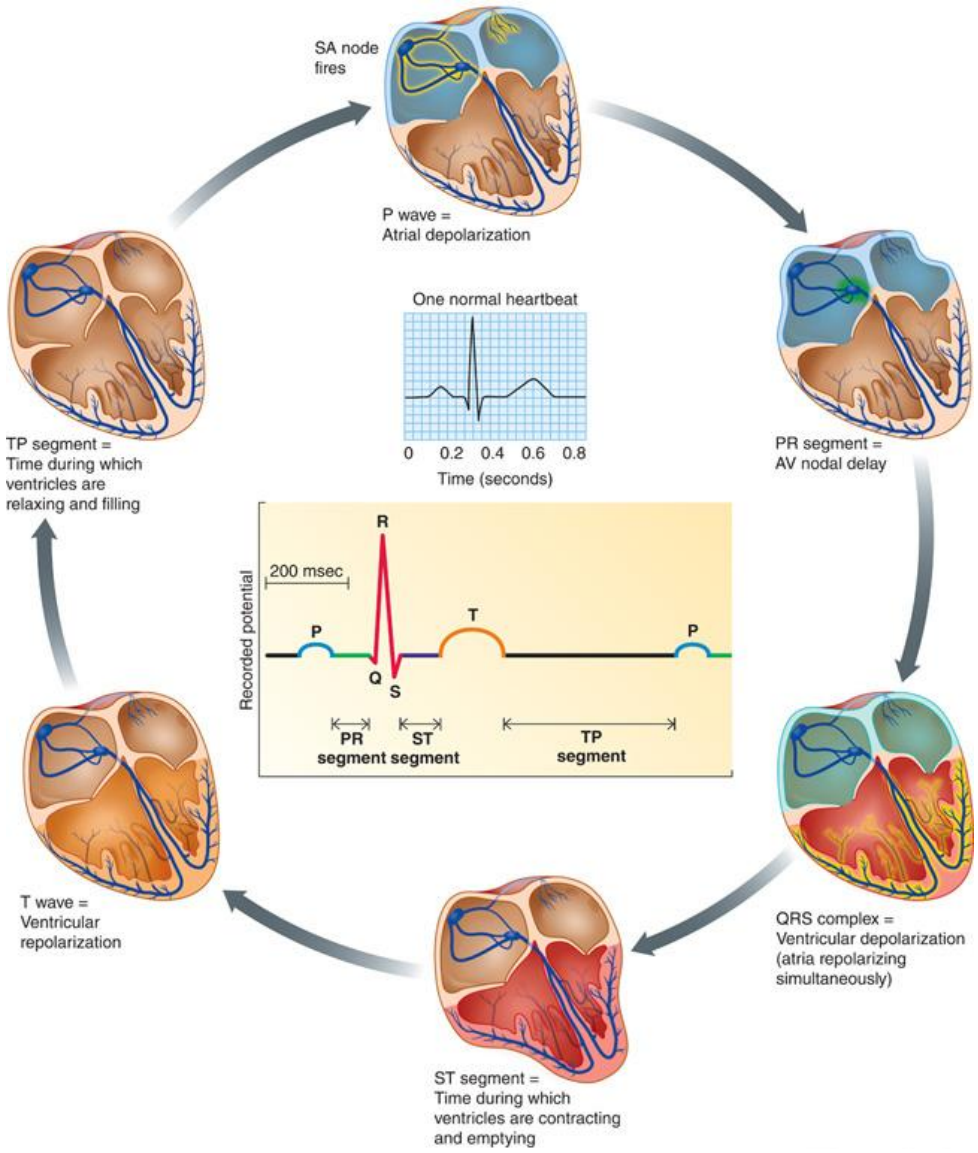
ECG (electrocardiogram)

- **ECG** is a record of overall spread of electrical activity through the heart
- ECG is a record of electrical activity present in body fluids generated by cardiac impulse that reaches the body surface, not a direct recording of actual electrical activity of the heart

- **Three distinct wave forms of ECG**
  - P wave: atrial depolarization
  - QRS complex: ventricular depolarization (atrial repolarization is hidden)
  - T wave: ventricular repolarization

- Three points showing no net current flow in the heart muscle
  - PR segment: time between the end of P and onset of QRS
    - Reflects conduction through the AV node (AV nodal delay)
  - ST segment: time between QRS and T
    - Represents the period when the ventricles are fully depolarized (plateau phase)
  - TP segment: time between the end of T and the next P wave
    - Represents the electrically silent period between heartbeats (true baseline)

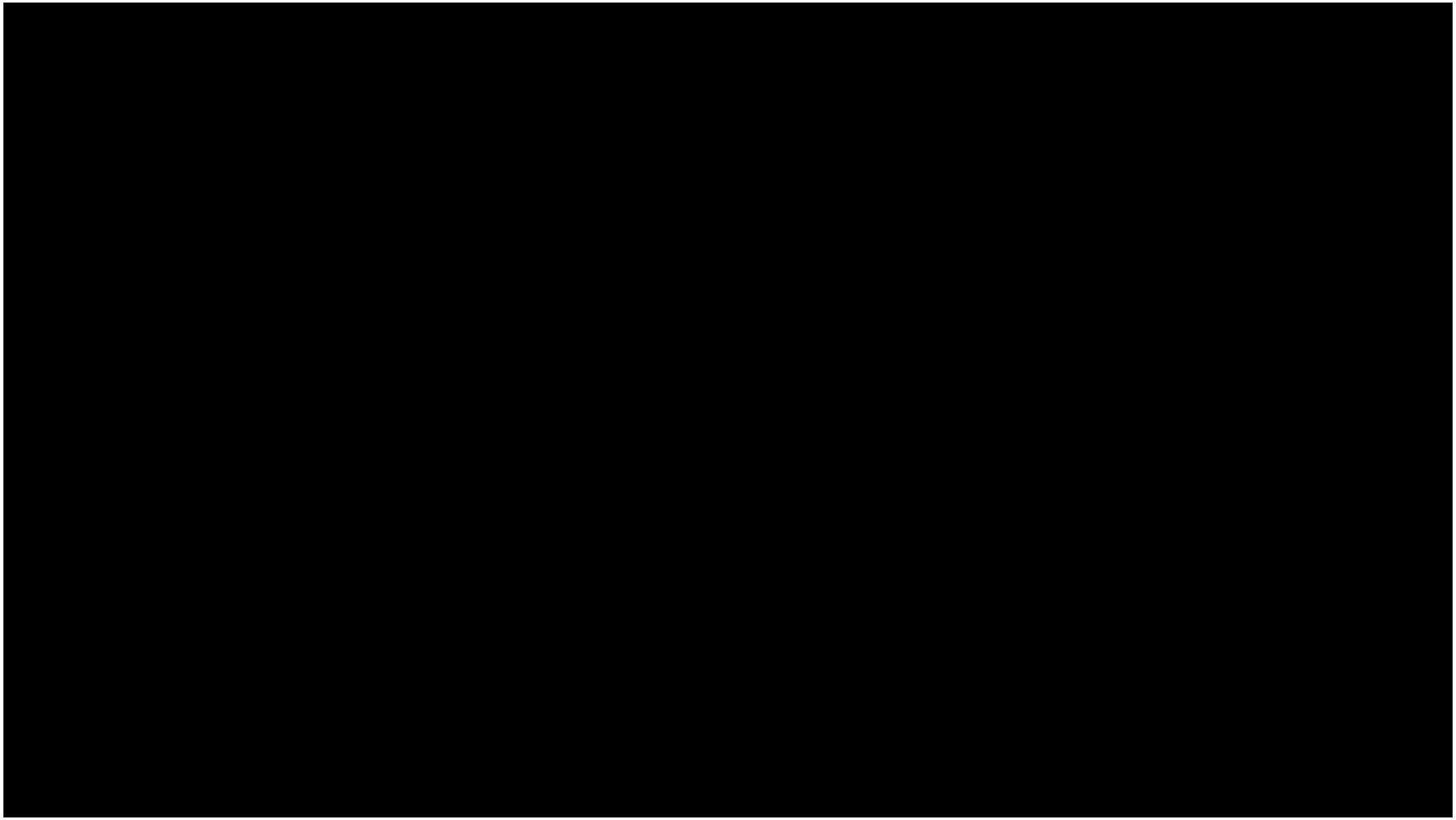
ECG (electrocardiogram) and cardiac cycle



- The ECG detect abnormal heart rates and rhythms and heart muscle damage
  - Abnormalities in rate
  - Abnormalities in rhythm
  - Cardiac myopathies

9.3 Mechanical Events of the Cardiac Cycle

Cardiac cycle



https://www.youtube.com/watch?v=_vZ0lefPg_0

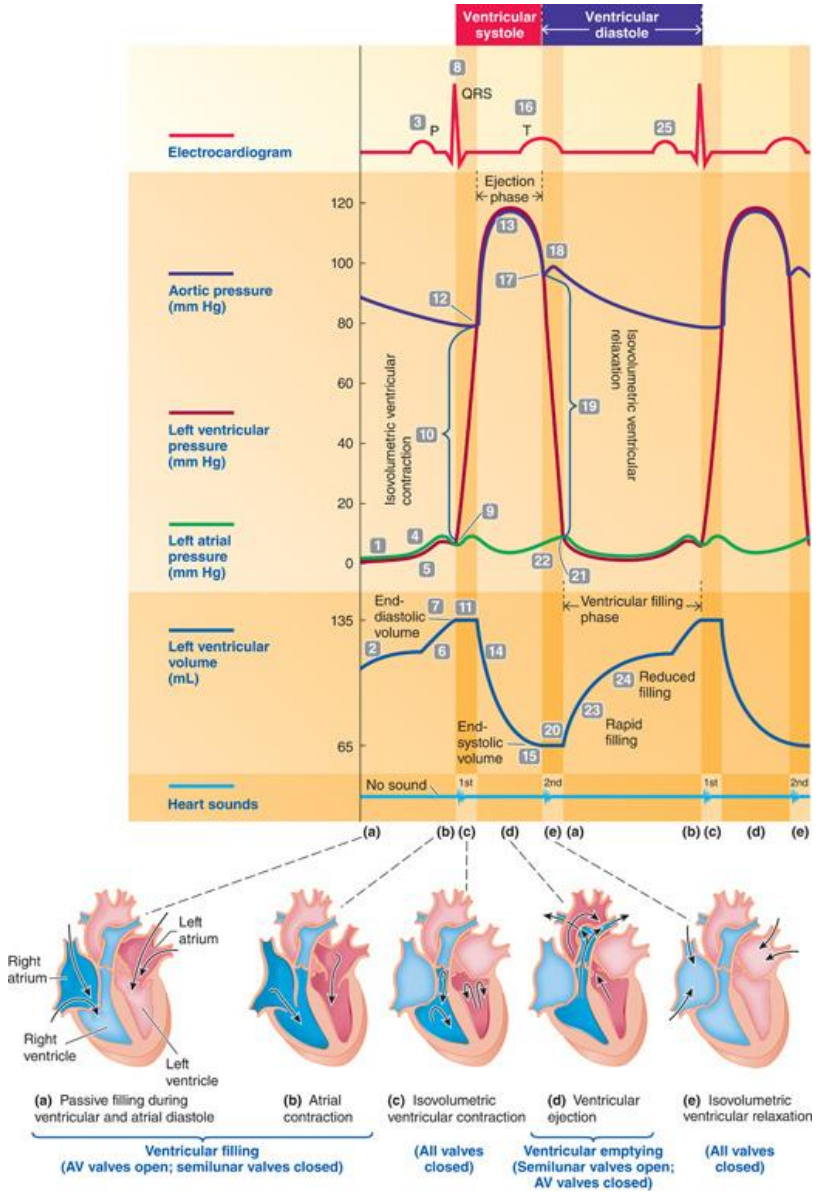
Cardiac cycle

- The heart alternatively contracts to empty and relaxes to fill
- Cardiac electrical activity -> contraction, relaxation -> blood flow changes

- Systole (수축기): contraction and emptying
- Diastole (이완기): relaxation and filling

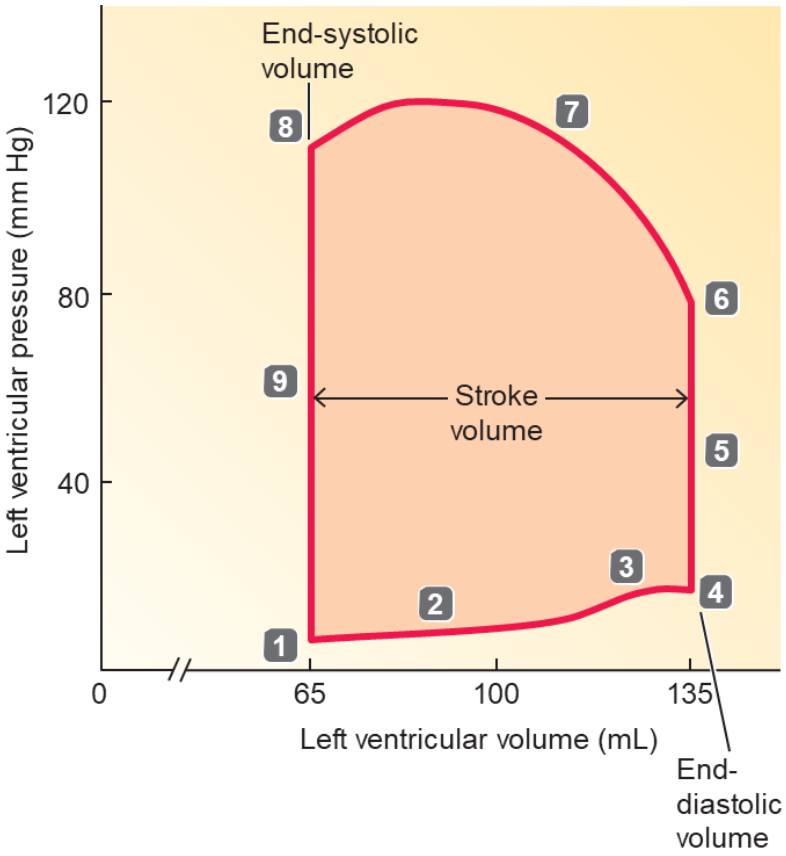
- Ventricular filling (심실 충만기): The ventricles are relaxed, allowing blood to flow from atria into ventricles (심실이 이완되면서 혈액이 심방에서 심실로)
- Isovolumetric contraction (등용적 수축기): The ventricles begin to contract, but all valves are closed, so ventricular volume remains unchanged (심실이 수축을 시작, 모든 판막이 닫혀 있어 심실 용적은 변하지 않음)
- Ventricular ejection (박출기): Ventricular pressure rises, opening the semilunar valves and ejecting blood into artery (심실 압력이 높아져 반월판 열리고 혈액이 동맥으로)
- Isovolumetric relaxation (등용적 이완기): The ventricles begin to relax, but all valves are closed, so ventricular volume remains unchanged (심실 이완 시작하지만 모든 판막이 닫혀 있어 심실 용적은 변하지 않음)

Cardiac cycle



- (a),(b) **Ventricular filling**
  - AV valves: open
  - Semilunar valves: closed
  - Blood flows from atria → ventricles
  - (late phase) atrial contraction completes filling
  - ECG: P wave, heart sound: X
- (c) **Isovolumetric contraction**
  - All valves: closed
  - Ventricles start contracting → pressure ↑ (volume change X)
  - ECG: QRS complex, heart sound: S1 (AV valves close)
- (d) **Ventricular ejection**
  - AV valves: closed
  - Semilunar valves: open
  - Blood ejected into aorta & pulmonary artery
  - ECG: (during ST segment), heart sound: X
- (e) **Isovolumetric relaxation**
  - All valves: closed
  - Ventricles relax → pressure ↓ (volume change X)
  - ECG: T wave, heart sound: S2 (semilunar valves close)

Cardiac cycle



1 The AV valve opens.

2 Passive ventricular filling occurs. Volume increases considerably and pressure increases slightly as blood enters.

3 Atrial contraction completes ventricular filling. End-diastolic volume is reached at the end of this phase.

4 The AV valve closes.

5 Isovolumetric ventricular contraction occurs. Volume remains constant; pressure increases markedly.

6 The aortic valve opens.

7 A stroke volume of blood is ejected. As blood leaves, volume decreases considerably as pressure peaks, then falls more slowly until end-systolic volume is reached at the end of this phase.

8 The aortic valve closes.

9 Isovolumetric ventricular relaxation occurs. Volume remains constant; pressure falls sharply.

Back to step 1.

9.4 Cardiac Output and its Control

Cardiac output and its control

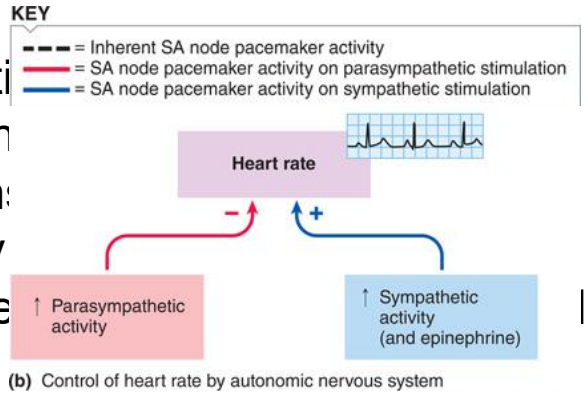
- Cardiac output (CO) is the volume of blood pumped by each ventricle per minute (not the total amount pumped by heart)
  - The blood volume flowing through pulmonary circulation is the same as that flows through the systemic circulation
  - Two determinants of CO is **heart rate** (HR) and **stroke volume** (SV)
- Heart rate is primarily determined by autonomic influences on the SA node (70 beats per minute, on average)
  - Heart is innervated by both divisions of autonomic nervous system, sympathetic and parasympathetic divisions
  - They modify the rate and strength of contraction, while nervous stimulation is unnecessary for initiating contraction
  - Vagus nerve, parasympathetic nerve reaches the atrium, especially SA and AV nodes
  - Cardiac sympathetic nerve reaches both the atrium and ventricle
  - Both nervous system regulates cAMP pathway in the innervated cardiac muscle cells

Cardiac output and its control

TABLE 9-1 Effects of the Autonomic Nervous System on Heart Activity

Area Affected	Effect of Parasympathetic Stimulation	Effect of Sympathetic Stimulation
SA node	Decreases the rate of depolarization to threshold; decreases the heart rate	Increases the rate of depolarization to threshold; increases the heart rate
AV node	Decreases excitability; increases the AV nodal delay	Increases excitability; decreases the AV nodal delay
Ventricular conduction pathway	No effect	Increases excitability; hastens conduction through the bundle of His and Purkinje cells
Atrial muscle	Decreases contractility; weakens contraction	Increases contractility; strengthens contraction
Ventricular muscle	No effect	Increases contractility; strengthens contraction
Adrenal medulla (an endocrine gland)	No effect	Promotes secretion of epinephrine, a hormone that augments sympathetic nervous system actions
Veins	No effect	Increases venous return, which increases the strength of cardiac contraction via intrinsic control

- ACh released from parasympathetic nerve binds to muscarinic cholinergic receptor and is coupled to inhibitory G protein that reduces activity of cAMP pathway
- NE released by sympathetic nerve binds to adrenergic receptor and is coupled to stimulatory G protein that activates cAMP pathway in muscle cells
- Activati
  phosph
  protein
  activity
  channe lls

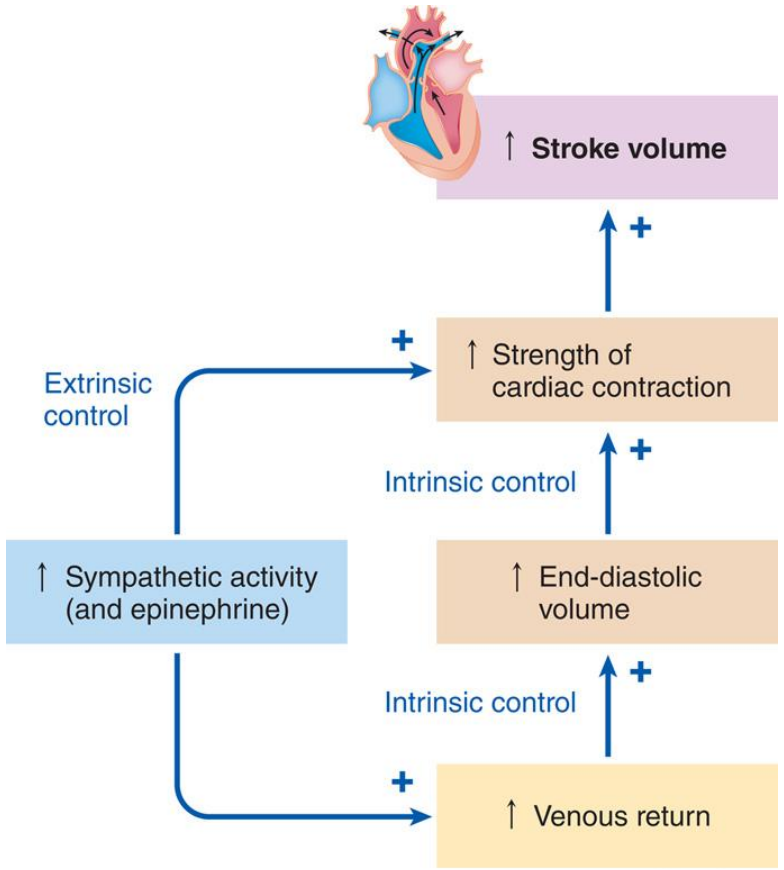


(b) Control of heart rate by autonomic nervous system

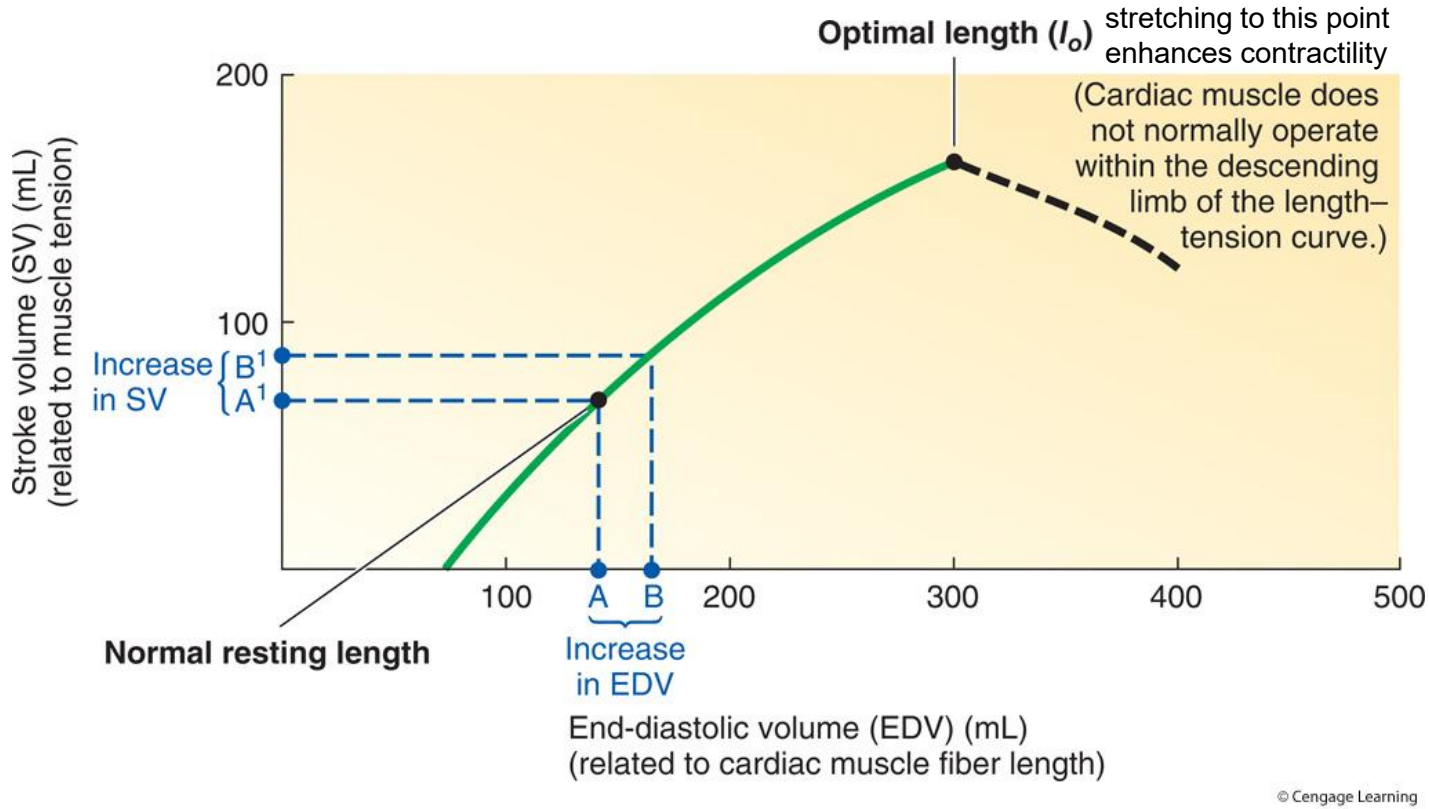
Stroke volume

- Stroke volume is determined by the extent of venous return and by sympathetic activity
- Two types of control influence SV
  - **Intrinsic control**: related to the extent of venous return
  - **Extrinsic control**: related to the extent of sympathetic stimulation of the heart
  - Both factors increase the strength of heart contraction

- Intrinsic control
  - Heart's inherent ability to regulate SV, depending on end-diastolic volume (EDV)
  - **Length-tension relationship**
  - Cardiac muscles are not attached to bones
  - The main determinant of muscle fiber length is the degree of diastolic filling
  - **Frank-Starling law of the heart**



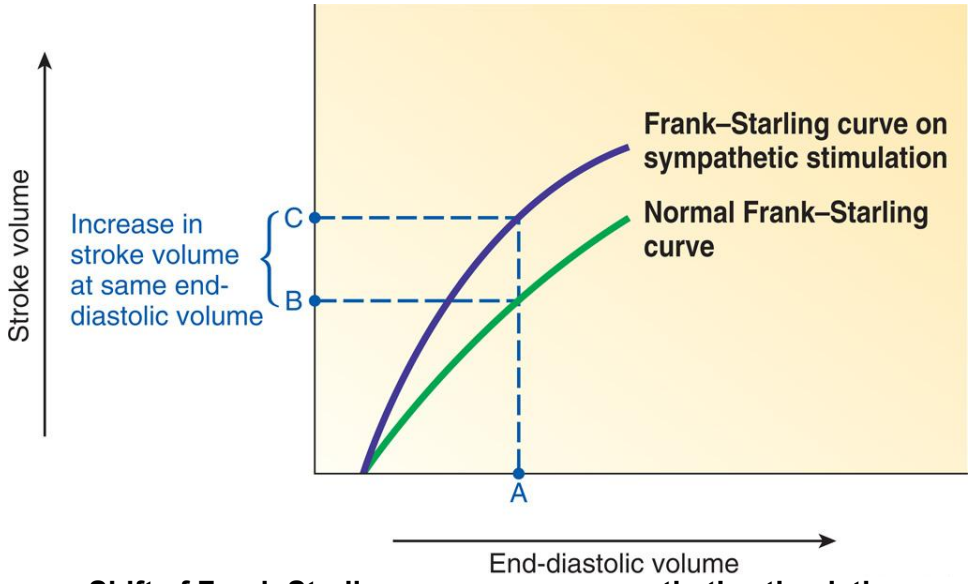
Frank-Starling law of the heart



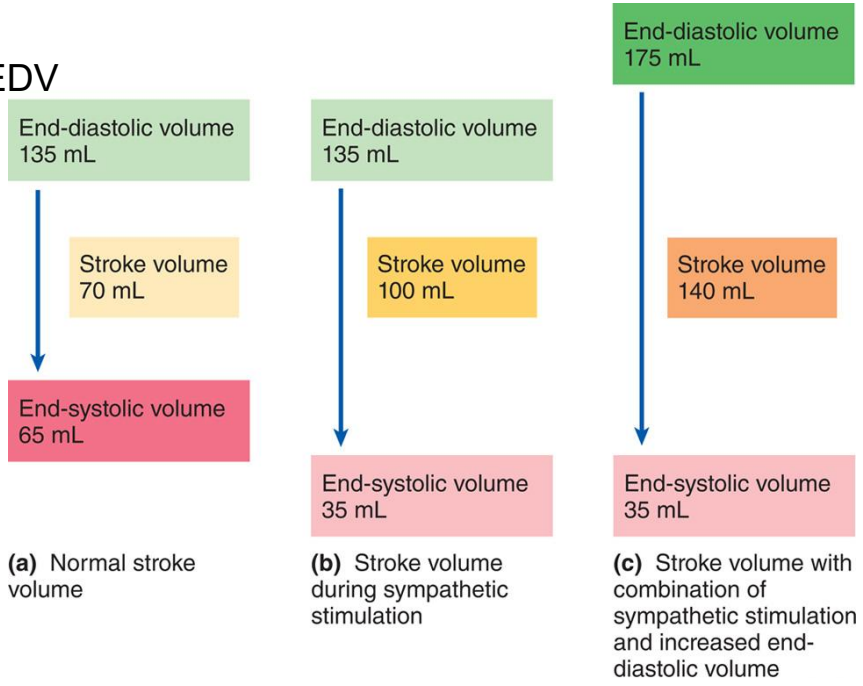
- Intrinsic relationship between EDV and SV
- Increased filling = Increased output
- Preload = the maximum stretching of the ventricles at the end of diastole just before contraction
- Sympathetic → venoconstriction -> venous return ↑→ preload ↑ → SV ↑

Stroke volume

- Extrinsic control
  - Sympathetic control increases **venous return**
  - Sympathetic stimulation enhance the heart's **contractility** via **epinephrine (E)** and **norepinephrine (NE)**
  - Contractility = strength of contraction at any given EDV
  - Increased Ca^{2+} influx by E and NE
  - Generate more force through cross-bridge cycling



Shift of Frank-Starling curve upon sympathetic stimulation

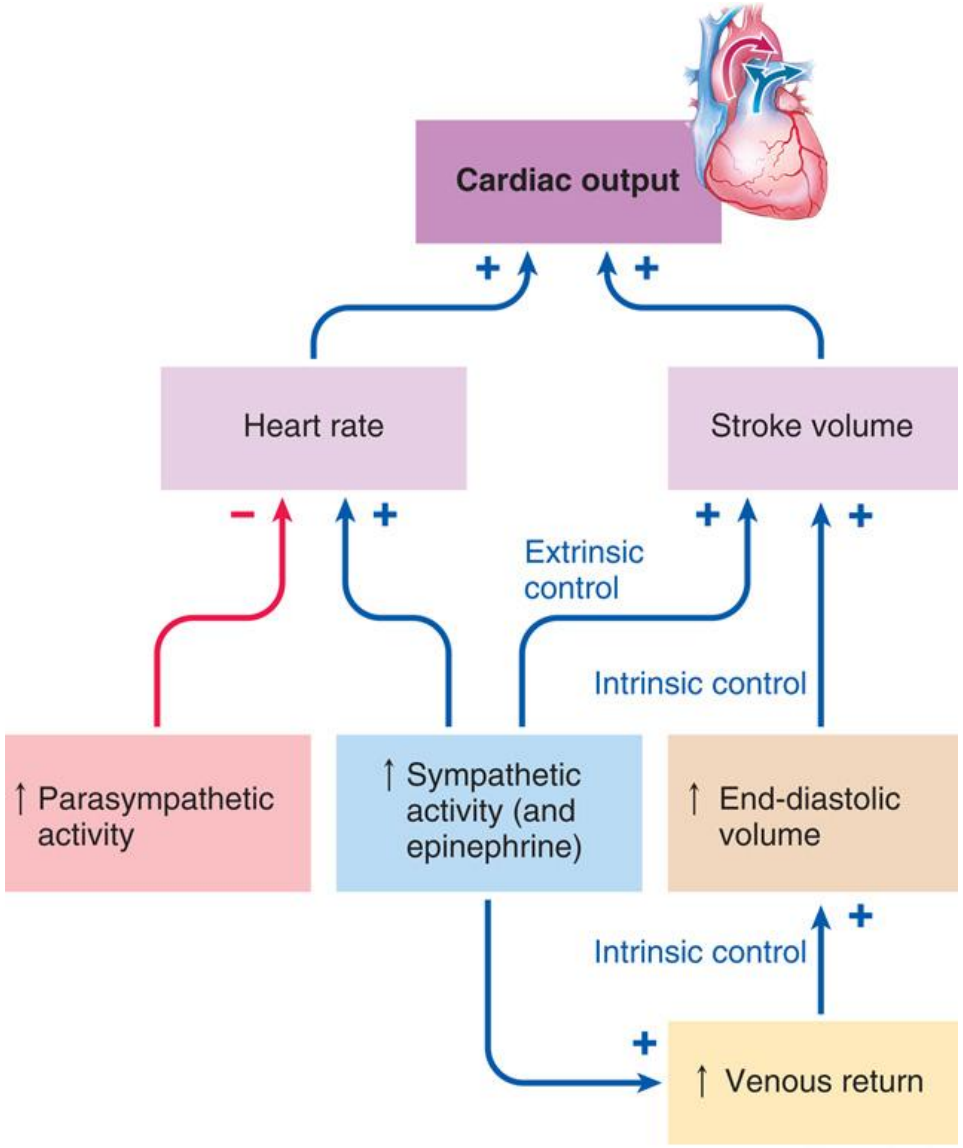


Effect of sympathetic stimulation on stroke volume

Ejection fraction = SV/EDV

51.9% 74.1% 80.0%

Control of cardiac output



- CO = HR x SV
- HR
  - Autonomic influences on the SA node
  - Para and sympathetic
- SV
  - Intrinsic control: Frank-Starling
  - Extrinsic control
    - Venous return
    - Contractility

9.5 Nourishing the Heart Muscle

Coronary circulation

- Heart receives most of its blood supply through the coronary circulation system during **diastole**
  - Cardiac muscle cells contain abundant mitochondria, depending on oxygen supply
  - Abundance of myoglobin storing limited amounts of oxygen
  - Heart cannot receive oxygen and nutrients from blood within chambers
    - Watertight endothelial lining blocking blood to pass into myocardium
    - Too thick heart wall for diffusion of supplies to the individual cardiac cells
- Coronary circulation
  - Heart cells receive blood through blood vessels
  - Blood flow to the heart muscle cells is reduced during systole
    - Contracting myocardium compresses coronary arteries
    - Opened aortic valve blocks entrance to the coronary vessels
  - 70% of coronary arterial flow occurs during diastole

Nutrient supply to the heart

- During exercise, extra blood is delivered by vasodilation, enlargement of coronary vessels, to supply blood during diastole
- When cardiac activity is increased, **adenosine** (formed from ATP) is released and acts as a paracrine, inducing dilation of coronary blood vessels
- The heart cannot get enough ATP from anaerobic metabolism
- Fuel sources for the heart include
  - Fatty acid: main source
  - Glucose and lactate
  - Quite tolerable and flexible to metabolite availability

Myocardial damage like myocardial infarction (heart attack)

